Second Lie Derivative on a Manifold

Theorem

Let M be a smooth manifold with local coordinates

$$(x^1, \dots, x^n),$$

let

$$h : M \to \mathbb{R}$$

be a smooth scalar function, and let

$$f = f^i \frac{\partial}{\partial x^i}$$

be a smooth vector field.

Let ∇ be a connection on M. Then

$$\boxed{L_f^2 h = \nabla^2 h(f, f) + dh(\nabla_f f)}$$

where $\nabla^2 h = \nabla dh$ is the covariant Hessian of h.

We derive this identity step by step.

1. Local coordinates and notation

Let

$$(x^1, \dots, x^n)$$

be local coordinates on M.

We use the notation

$$\partial_i = \frac{\partial}{\partial x^i}.$$

The vector field f is written as

$$f = f^i \partial_i.$$

Here $f^i = f^i(x)$ are the coordinate components of f.

The scalar function h is

$$h = h(x^1, \dots, x^n).$$

Its differential is the covector

$$dh = (\partial_i h)\, dx^i.$$

Throughout the derivation, repeated indices are summed according to the Einstein summation convention.

2. First Lie derivative

For a scalar function h, the Lie derivative along the vector field f is simply the directional derivative:

$$L_f h = f(h).$$

Since

$$f = f^i \partial_i,$$

we obtain

$$L_f h = f^i \partial_i h.$$

Thus

$$\boxed{L_f h = f^i \partial_i h.}$$

We can also express this using the differential dh.
Since

$$dh = (\partial_i h)\, dx^i,$$

we have

$$dh(f) = (\partial_i h)\, dx^i (f^j \partial_j).$$

Using

$$dx^i(\partial_j) = \delta^i_j,$$

we obtain

$$dh(f) = (\partial_i h) f^i.$$

Therefore

$$\boxed{L_f h = dh(f) = f^i \partial_i h.}$$

3. Second Lie derivative

Now apply L_f once more:

$$L_f^2 h = L_f(L_f h).$$

Since $L_f h$ is a scalar function,

$$L_f^2 h = f^j \partial_j (L_f h).$$

Substituting

$$L_f h = f^i \partial_i h,$$

we obtain

$$L_f^2 h = f^j \partial_j \left(f^i \partial_i h \right).$$

Therefore,

$$\boxed{L_f^2 h = f^j \partial_j \left(f^i \partial_i h \right).}$$

4. Expand using the ordinary product rule

Apply the ordinary product rule:

$$\partial_j \left(f^i \partial_i h \right) = (\partial_j f^i)(\partial_i h) + f^i \partial_j \partial_i h.$$

Multiplying by f^j,

$$L_f^2 h = f^j (\partial_j f^i)(\partial_i h) + f^j f^i \partial_j \partial_i h.$$

Thus

$$\boxed{L_f^2 h = f^j f^i \partial_j \partial_i h + f^j (\partial_j f^i) \partial_i h.}$$

At this stage we have used only ordinary coordinate derivatives.
There are two contributions:

$$f^j f^i \partial_j \partial_i h$$

is the ordinary second derivative of h, while

$$f^j (\partial_j f^i) \partial_i h$$

contains the change of the vector field f along itself.
The second term is not yet geometrically invariant because

$$\partial_j f^i$$

does not transform as a tensor.

This motivates the introduction of a connection.

5. Introduce a connection

Let ∇ be a connection on M.

We want to calculate the covariant derivative of f in the direction f:

$$\nabla_f f.$$

Since

$$f = f^j \partial_j,$$

linearity in the first argument gives

$$\nabla_f f = \nabla_{f^j \partial_j} f.$$

Therefore

$$\boxed{\nabla_f f = f^j \nabla_{\partial_j} f.}$$

Now write the vector field being differentiated as

$$f = f^i \partial_i.$$

Hence

$$\nabla_{\partial_j} f = \nabla_{\partial_j} \left(f^i \partial_i \right).$$

6. Product rule for the connection

The connection satisfies the Leibniz rule:

$$\nabla_X(\phi Y) = X(\phi)Y + \phi \nabla_X Y.$$

Therefore,

$$\nabla_{\partial_j} \left(f^i \partial_i \right) = (\partial_j f^i)\partial_i + f^i \nabla_{\partial_j} \partial_i.$$

Substituting into $\nabla_f f$,

$$\nabla_f f = f^j \left[(\partial_j f^i)\partial_i + f^i \nabla_{\partial_j} \partial_i \right].$$

Hence

$$\boxed{\nabla_f f = f^j (\partial_j f^i)\partial_i + f^j f^i \nabla_{\partial_j} \partial_i.}$$

7. Introduce the Christoffel symbols

The Christoffel symbols of the connection are defined by

$$\boxed{\nabla_{\partial_j}\partial_i = \Gamma^k{}_{ji}\partial_k.}$$

Substituting this into the previous expression gives

$$\nabla_f f = f^j(\partial_j f^i)\partial_i + f^j f^i \Gamma^k{}_{ji}\partial_k.$$

We want to express the result using one common free index.
Rename the dummy indices in the second term:

$$i \to j, \qquad j \to k, \qquad k \to i.$$

Then

$$f^j f^i \Gamma^k{}_{ji}\partial_k = \Gamma^i{}_{jk} f^j f^k \partial_i.$$

Consequently,

$$\nabla_f f = \left[f^j \partial_j f^i + \Gamma^i{}_{jk} f^j f^k\right]\partial_i.$$

Therefore the components of the covariant acceleration are

$$\boxed{(\nabla_f f)^i = f^j \partial_j f^i + \Gamma^i{}_{jk} f^j f^k.}$$

Solving for the ordinary derivative gives

$$\boxed{f^j \partial_j f^i = (\nabla_f f)^i - \Gamma^i{}_{jk} f^j f^k.}$$

8. Substitute the covariant acceleration

Recall from Section 4 that

$$L_f^2 h = f^j f^i \partial_j \partial_i h + f^j(\partial_j f^i)\partial_i h.$$

Using

$$f^j \partial_j f^i = (\nabla_f f)^i - \Gamma^i{}_{jk} f^j f^k,$$

we obtain

$$L_f^2 h = f^j f^i \partial_j \partial_i h + \left[(\nabla_f f)^i - \Gamma^i{}_{jk} f^j f^k\right]\partial_i h.$$

Expanding,

$$L_f^2 h = f^j f^i \partial_j \partial_i h - \Gamma^i{}_{jk} f^j f^k \partial_i h + (\nabla_f f)^i \partial_i h.$$

Rename the dummy indices in the Christoffel term. Then

$$\boxed{L_f^2 h = f^j f^i \left(\partial_j \partial_i h - \Gamma^k{}_{ji} \partial_k h\right) + (\nabla_f f)^k \partial_k h.}$$

This is the key intermediate expression.

9. The covariant derivative of a covector

The differential dh is a covector field:

$$dh = (\partial_i h)\, dx^i.$$

For a general covector field

$$\alpha = \alpha_i\, dx^i,$$

its covariant derivative has components

$$\boxed{(\nabla_j \alpha)_i = \partial_j \alpha_i - \Gamma^k{}_{ji} \alpha_k.}$$

Now choose

$$\alpha = dh.$$

Since

$$\alpha_i = \partial_i h,$$

we obtain

$$(\nabla_j dh)_i = \partial_j (\partial_i h) - \Gamma^k{}_{ji} \partial_k h.$$

Therefore,

$$\boxed{(\nabla_j dh)_i = \partial_j \partial_i h - \Gamma^k{}_{ji} \partial_k h.}$$

10. Identify the Hessian

The covariant Hessian of h is defined by

$$\boxed{\nabla^2 h = \nabla dh.}$$

Therefore its coordinate components are

$$\boxed{(\nabla^2 h)_{ji} = \partial_j \partial_i h - \Gamma^k{}_{ji} \partial_k h.}$$

Substituting this into the expression obtained in Section 8,

$$L_f^2 h = f^j f^i (\nabla^2 h)_{ji} + (\nabla_f f)^k \partial_k h.$$

The first term is precisely the Hessian evaluated on f and f:

$$\boxed{\nabla^2 h(f,f) = (\nabla^2 h)_{ji} f^j f^i.}$$

Thus,

$$L_f^2 h = \nabla^2 h(f,f) + (\nabla_f f)^k \partial_k h.$$

11. Identify the remaining term

Since

$$dh = (\partial_k h)\, dx^k,$$

the action of dh on the vector $\nabla_f f$ is

$$dh(\nabla_f f) = (\partial_k h)(\nabla_f f)^k.$$

Therefore,

$$\boxed{(\nabla_f f)^k \partial_k h = dh(\nabla_f f).}$$

Substituting this result gives the final formula.

12. Final geometric identity

We have derived

$$\boxed{L_f^2 h = \nabla^2 h(f,f) + dh(\nabla_f f).}$$

This is the desired coordinate-free identity.

13. Interpretation of the two terms

The formula

$$L_f^2 h = \nabla^2 h(f,f) + dh(\nabla_f f)$$

contains two fundamentally different contributions.

The first term,

$$\boxed{\nabla^2 h(f,f),}$$

describes the intrinsic second-order variation, or curvature, of the scalar function h in the direction f.

The second term,

$$\boxed{dh(\nabla_f f),}$$

describes the contribution caused by the covariant acceleration of the trajectory generated by f.

Thus,

$$\boxed{\text{second change of } h = \text{curvature of } h + \text{effect of trajectory acceleration.}}$$

14. Geodesic case

Suppose the integral curves of f are geodesics.

The geodesic condition is

$$\boxed{\nabla_f f = 0.}$$

In coordinates this becomes

$$\boxed{f^j \partial_j f^i + \Gamma^i{}_{jk} f^j f^k = 0.}$$

Therefore,

$$dh(\nabla_f f) = 0.$$

The general identity consequently reduces to

$$\boxed{L_f^2 h = \nabla^2 h(f, f).}$$

Thus, along a geodesic flow, the second Lie derivative of h is exactly the Hessian quadratic form in the direction f.

15. Geometric meaning

Let $\gamma(t)$ be an integral curve of f:

$$\dot{\gamma}(t) = f(\gamma(t)).$$

Then

$$\frac{d}{dt} h(\gamma(t)) = L_f h.$$

Differentiating once more,

$$\frac{d^2}{dt^2} h(\gamma(t)) = L_f^2 h.$$

Therefore,

$$\boxed{\frac{d^2}{dt^2}h(\gamma(t)) = \nabla^2 h(f,f) + dh(\nabla_f f).}$$

If $\gamma(t)$ is a geodesic,

$$\nabla_{\dot\gamma}\dot\gamma = 0,$$

and hence

$$\boxed{\frac{d^2}{dt^2}h(\gamma(t)) = \nabla^2 h(\dot\gamma,\dot\gamma).}$$

This is the fundamental relation between the Hessian and the second derivative of a function along a geodesic.

16. Summary of the derivation

The complete chain is

$$L_f h = f^i\partial_i h,$$

$$L_f^2 h = f^j\partial_j(f^i\partial_i h),$$

$$L_f^2 h = f^j f^i\partial_j\partial_i h + f^j(\partial_j f^i)\partial_i h,$$

$$(\nabla_f f)^i = f^j\partial_j f^i + \Gamma^i{}_{jk} f^j f^k,$$

$$f^j\partial_j f^i = (\nabla_f f)^i - \Gamma^i{}_{jk} f^j f^k,$$

$$L_f^2 h = f^j f^i\left(\partial_j\partial_i h - \Gamma^k{}_{ji}\partial_k h\right) + (\nabla_f f)^k\partial_k h,$$

$$(\nabla^2 h)_{ji} = \partial_j\partial_i h - \Gamma^k{}_{ji}\partial_k h,$$

and therefore

$$\boxed{L_f^2 h = \nabla^2 h(f,f) + dh(\nabla_f f).}$$

For a geodesic vector field,

$$\nabla_f f = 0,$$

so

$$\boxed{L_f^2 h = \nabla^2 h(f,f).}$$